CHM2242 Organic Chemistry – Typed Lecture Notes

Oct 31, 2026 · @Abdullahi Idris

1. Alcohols

Structure, properties and reactions of alcohol (notes dated 28/09/26). R = aryl, alkyl or cyclic.

Course outline (as listed in the notes)

1. Alcohols, ethers, epoxides
2. Aldehydes and ketones
3. Carboxylic acids and their derivatives
4. Amines
5. Introduction to α,β-unsaturated carbonyl compounds

General formula and classification

Alcohols have the general formula **R–OH** (R = alkyl group).

Class	Structure	Note
Methyl alcohol	CH_3–OH	
1° alcohol	R–CH_2–OH	
2° alcohol	R_2CH–OH (R and R′ on the carbinol carbon)	
3° alcohol	R_3C–OH	

Classification examples drawn in the notes

- 1-Methylcyclopentanol (CH_3 and OH on the same ring carbon): 3°
- Cyclohexanol (ring carbon bearing H and OH): 2°
- Ring–CH_2OH (benzyl-type alcohol, drawn as a ring with CH_2OH): 1°
- Cyclopentanol: 2°

- CH_2=CH–OH: written as 2° in the notes (the carbon bearing OH is bonded to only one other carbon, so check this one against your lecturer's answer)
- C_3H…–$C(CH_3)(OH)$–CH_3: 3° (the alkyl chain is partly cut off in the notes)

Isomerism

- C_2H_6O: CH_3CH_2OH (ethanol) and CH_3–O–CH_3 (dimethyl ether)
- C_3H_6O: CH_2=CH–CH_2–OH, CH_2=C(OH)–CH_3, and CH(OH)=CH–CH_3
- Also noted at the top of the next page (C_3H_6O isomers): cyclopropanol (a three-membered ring bearing OH), CH_3CH_2–CHO and CH_3–CO–CH_3

Physical properties

- Water and alcohols have similar physical properties.
- Alcohols are soluble in water.
- The presence of the –OH group makes them polar.
- The next point begins "Lo…" and is cut off at the bottom of the page (probably "Low…"; it is not finished in the notes).

Preparation of alcohols

1) Hydration of alkenes

```
C=C  +  H2O  --(OH⁻, H⁺)-->  –C(H)–C(OH)–
```

a. Acid-catalysed hydration of an alkene (involves protonation)

```
C=C  +  H2O  --(H⁺)-->  –C(H)–C(OH)–
```

Mechanism:

1. Protonation of the double bond to form a carbocation: C=C + H–O⁺(H)–H → H–C–C⁺ + H_2O
2. Nucleophilic attack by water on the carbocation gives a protonated alcohol: –C⁺ + $H_2\ddot{O}$: → –C–O⁺(H)–H
3. Deprotonation gives the alcohol: –C(H)–C–$\ddot{O}^+H_2$ + H_2O ⇌ –C(H)–C(OH)– + H_3O^+

b. Hydration by oxymercuration–demercuration

```
C=C  +  Hg(OAc)2, H2O  -->  –C(OH)–C(HgOAc)–  -->  –C(OH)–C(H)–
```

Mercuric acetate: $Hg(OAc)_2$ = $Hg(O_2CCH_3)_2$. (The second-step reagent is not written on the page; it is the reduction that replaces HgOAc with H, $NaBH_4$ as in the ether notes.)

c. Hydroboration

```
CH3–C(CH3)=CH–CH3  --(B2H6)-->  CH3–CH(CH3)–CH(BH2)–CH3  (an alkylborane)  --(H2O2, OH⁻)-->  CH3–CH(CH3)–CH(OH)–CH3
```

Product: 3-methylbutan-2-ol.

2) Formation of alcohols from aldehydes and ketones using Grignard reagents (GR)

Grignard reagents: R–Mg–X, for example CH_3–Mg–Cl and C_2H_5–MgBr.

Carbonyl compound	Reaction	Product
Formaldehyde, H–CHO	H–C(=O)–H + R–MgX → …	R–CH_2–OH (1°)
Aldehyde (other than formaldehyde)	R–C(=O)–H + R′MgX → …	R–CH(R′)–OH (2°)
Ketone	R–C(=O)–R + R′MgX → …	R–C(R′)(R)–OH (3°)

Example: C_6H_5–MgBr + $(CH_3)_2C{=}O$, in ether → C_6H_5–$C(CH_3)_2$–OH

Exercise given in the notes: show how the following can be prepared

1. Cyclohexyl–CH_2OH (hexagon ring bearing CH_2OH)
2. Cyclopentyl–CH_2OH (pentagon ring bearing CH_2OH)
3. A branched primary alcohol drawn as a skeleton: a methyl-branched chain ending in –OH
4. Cyclopentyl–$C(CH_3)(OH)$–CH_2OH (ring carbon attached to a carbon bearing CH_3, OH and CH_2OH)
5. 3-Phenylhexan-3-ol

3) Reduction of carbonyl compounds with hydride reagents ($NaBH_4$, $LiAlH_4$)

```
C=O  --(NaBH4 or LiAlH4)-->  –CH–OH
```

4) Alcohols from hydrolysis of an ester

```
R–C(=O)–OR  +  H2O  -->  R–OH  +  R–COO⁻  (carboxylate)
```

Example (as written): $CH_3CH_2–C(=O)–CH_2CH_3 + Na^+OH^- \rightarrow CH_3CH_2–OH + CH_3CH_2–C(=O)–O^-Na^+$. The ester oxygen is missing from the starting material in the notes; it should read $CH_3CH_2–C(=O)–O–CH_2CH_3$ (ethyl propanoate). A margin note on the page says "ether" next to the ester.

2. Reactions of alcohols (R–OH)

Generally, R–OH undergo substitution (SN1, SN2), elimination (E1, E2), esterification, reduction and oxidation reactions, as well as formation of ethers.

Substitution reactions of R–OH

1. Conversion of R–OH to alkyl halides

The most commonly used reagents for this conversion are:

- Hydrogen halides (HCl, HBr, HI)
- Phosphorus tribromide (PBr_3)
- Thionyl chloride ($SOCl_2$)

a) Reaction of R–OH with HX

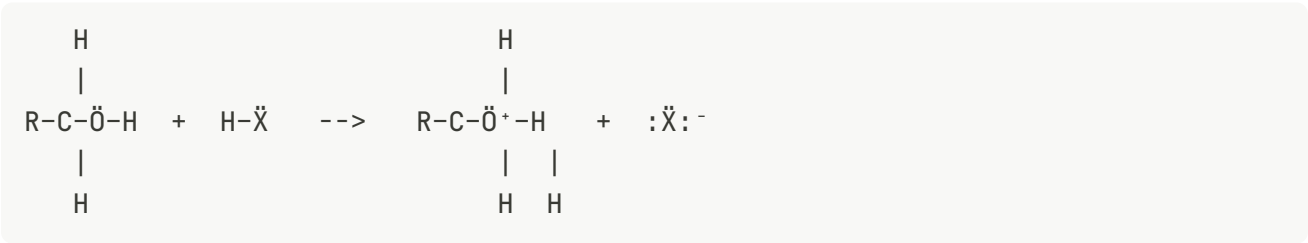
```
R–OH  +  HX  -->  R–X  +  H2O
```

Note: 1° alcohols and MeOH react to form alkyl halides under acidic conditions by the SN2 mechanism.

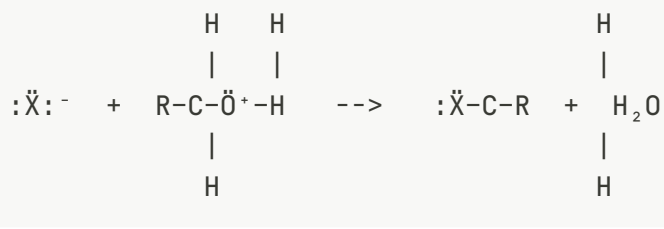
Example from the earlier notes (5/10/2025): $CH_3–CH(OH)–CH_2–CH_3 + HI \rightarrow CH_3–CH(I)–CH_2–CH_3 + H_2O$.

Mechanism:

- **Step 1: protonation of the alcohol by an acid.**

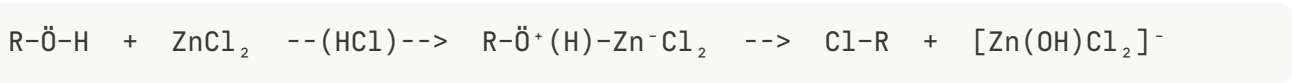


- **Step 2: displacement of the poor leaving group (H_2O) by the halide ion.**



where X = Br, I.

However, hydrogen chloride does not react with 1° or 2° alcohols unless $ZnCl_2$ or some other Lewis acid is added to the reaction mixture:



b) Reaction of R–OH with phosphorus tribromide (PBr_3)

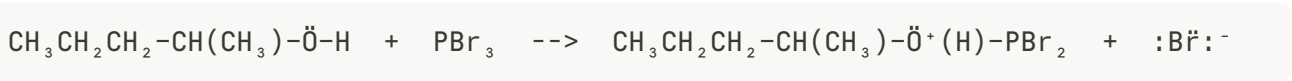


Example: reaction of (R)-2-pentanol with PBr_3.

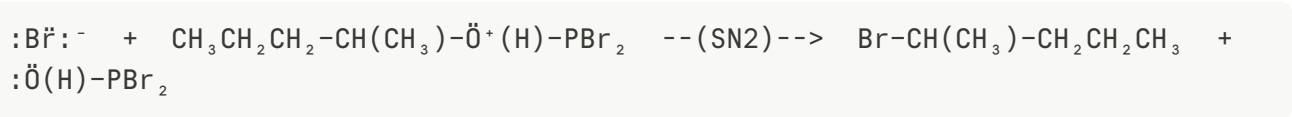


Mechanism:

- **Step 1: displacement of bromide and formation of the leaving group.** The oxygen lone pair attacks phosphorus and bromide leaves:



- **Step 2: bromide displaces the leaving group to give (S)-2-bromopentane** (SN2, back-side attack):



Notes:

1. Rearrangement is uncommon because no carbocation is formed.
2. This mechanism is SN2 displacement where Br^- attacks the back side of the alkyl group. The attack is hindered if the alkyl group is tertiary.

c) Reaction of R–OH with thionyl chloride ($SOCl_2$)

(The notes write the formula as $SOCl_3$ in this heading; it is $SOCl_2$, as written in the reagent list.)

```
R–OH  +  Cl–S(=O)–Cl  --(heat)-->  R–Cl  +  SO2  +  HCl
```

Mechanism:

1. The alcohol oxygen attacks sulfur: R–Ö–H + $S(=O)Cl_2$ → R–Ö⁺(H)–S(=O)Cl + Cl^-
2. Loss of H^+ (as HCl) gives the alkyl chlorosulfite: R–O–S(=O)–Cl + HCl
3. Chloride attacks carbon in the fast step, releasing SO_2: R–O–S(=O)Cl → R–Cl + SO_2 (the notes draw it as R^+ and $^-$O–S(=O)Cl collapsing to R–Cl + S=O)

Dehydration of R–OH

```
  H  OH
  |  |
–C--C–   ⇌   C=C  +  H2O
  |  |
```

Dehydration follows **Zaitsev's rule**: if two or more alkenes might be formed by deprotonation of the carbocation, the most highly substituted alkene predominates.

Example (2-butanol, dehydrated; the minor product is cut off at the page edge):

```
CH3–CH(OH)–CH2–CH3  -->  CH3–CH=CH–CH3  +  CH2=CH–CH2–CH3
                         2-butene             1-butene
                         major (70%)          minor (30%)
```

3. Ethers

Ethers are organic compounds of the general formula **R–O–R**, **Ar–O–R** or **Ar–O–Ar**, where Ar is phenyl or some other aromatic group.

Naming of ethers

1) Name the two groups attached to oxygen, and follow these names with the word "ether". Use alphabetical order if the two alkyl groups are not identical.

Structure (as drawn)	Name
$CH_3CH_2–O–CH_2CH_3$	Diethyl ether
$(CH_3)_2CH–O–CH(CH_3)_2$	Diisopropyl ether
$(CH_3)_3C–O–CH_3$	tert-Butyl methyl ether
$CH_3CH_2CH(CH_3)–O–CH_3$	2-Butyl methyl ether (sec-butyl methyl ether)

2) Name as alkoxy followed by the alkane name of the longest alkyl group.

Structure (as drawn)	Name
$CH_3CH_2–O–CH_2CH_3$	Ethoxyethane
$CH_3CH_2–O–CH_2CH_2CH_2CH_3$	Ethoxybutane
$CH_3CH_2CH(OCH_3)CH_2CH_2CH_3$	3-Methoxyhexane

Symmetrical ethers contain two identical alkyl groups, while unsymmetrical ethers have two different alkyl groups.

Preparation of ethers

1) Williamson synthesis: treatment of an alkyl halide with a sodium alkoxide.

```
R–X  +  R'O⁻Na⁺   -->  R–O–R'
R–X  +  ArO⁻Na⁺   -->  R–O–Ar
```

Yield from R–X: CH_3 > 1° > 2° > 3°.

Example: phenol + CH_3CH_2Br, aq. NaOH → ethoxybenzene ($C_6H_5–O–CH_2CH_3$)

2) Alkoxymercuration–demercuration

```
C=C  +  ROH  +  Hg(OOCCF₃)₂  -->  –C(OR)–C(HgOOCCF₃)–  --(NaBH₄)-->  –C(OR)–C(H)–
```

The reaction follows Markovnikov's rule.

Reactions of ethers

Ethers are very unreactive, hence they are used as solvents. However, they are cleaved by acids to form R–X.

Acid-catalysed cleavage of ethers

```
R–OH  +  HX  -->  R–X  +  H2O          (alcohol + hydrogen halide → alkyl
halide + water)
R–O–R'  +  HX  -->  R–X  +  R'–OH
```

The cleavage is carried out with excess hydrogen halide and heat:

```
R–O–R'  +  2 HX  --(heat)-->  R–X  +  R'–X  +  H2O

CH3CH(OCH3)CH2CH3  --(HBr, heat)-->  CH3CH(Br)CH2CH3  +  CH3Br
```

Reactivity of HX: **HI > HBr >> HCl**

Mechanism of the reaction

```
CH3CH2–O–CH2CH3  +  HBr  --(heat)-->  2 CH3CH2Br  +  H2O
```

1. **Protonation of oxygen (fast, reversible):**

```
CH3CH2–Ö–CH2CH3  +  H–Br   ⇌   CH3CH2–O+(H)–CH2CH3  +  Br-
                                  diethyloxonium ion
```

2. **Nucleophilic attack of the halide anion on a carbon of the dialkyloxonium ion (slow):**

```
Br-  +  CH3CH2–O+(H)–CH2CH3   --(slow)-->  CH3CH2Br  +  CH3CH2OH
```

3. **Third and fourth steps** (the alcohol formed is protonated, then attacked by bromide):

```
CH3CH2OH  +  HBr   ⇌ (fast)   CH3CH2–O+H2  +  Br-
Br-  +  CH3CH2–O+H2   --(slow)-->  CH3CH2Br  +  H2O
```

Crown ethers

Crown ethers are cyclic polyethers containing repeating $-OCH_2CH_2-$ units. They look cumbersome but are actually pretty simple to name. They are named **x-crown-y**, where x is the total number of carbon and oxygen atoms in the cyclic backbone and y is the number of oxygen atoms.

Name	Ring atoms (x)	Oxygen atoms (y)
12-crown-4	12	4
14-crown-4	14	4
15-crown-5	15	5

Crown ethers have valuable synthetic advantages. They form stable metal complexes by binding to small metal cations such as K^+, Na^+, etc., using the oxygen atoms as donors. In other words, the metal cations are held in the cavity of the crown ether.

4. Epoxides

Epoxides are cyclic ethers with a three-membered ring. The structure of an epoxide has an oxygen atom attached to two adjacent carbon atoms.

```
      O
     / \          (drawn in the notes: the basic epoxide ring, a methyl-
substituted epoxide,
    C---C          and epoxides fused to cyclobutane and cyclohexane rings)
```

Naming of epoxides

- Consider the longest carbon chain as the parent name.
- Count from the side nearest to oxygen.
- Specify the positions of the carbon atoms bearing the oxygen using the numbers 1, 2, …
- Add the word "epoxy" after the numbers.

Structure	Name
Three-membered ring, O on C1 and C2 of a two-carbon chain	1,2-Epoxyethane
Epoxide on C2 and C3 of a four-carbon chain	2,3-Epoxybutane
Epoxide fused to a cyclobutane ring	1,2-Epoxycyclobutane

Preparation of epoxides

1) Ethylene oxide (1,2-epoxyethane) is mainly prepared by exposing an alkene (ethene) to oxygen in the presence of silver oxide as a catalyst at high temperature.

```
CH2=CH2  --(O2, Ag2O, 300 °C)-->  ethylene oxide (three-membered ring C–C–O)
```

2) Epoxides are prepared by treatment of an alkene with a peroxyacid (RCO_3H), typically m-chloroperoxybenzoic acid.

```
cyclohexene  +  m-chloroperoxybenzoic acid   --(CH2Cl2 solvent)-->  1,2-
epoxycyclohexane  +  m-chlorobenzoic acid
```

(m-Chloroperoxybenzoic acid: a benzene ring with Cl at the meta position and a –C(=O)–O–O–H group. m-Chlorobenzoic acid: the same ring with –C(=O)–OH.)

3) Treatment of a halohydrin with a base.

```
C=C  +  X2  --(H2O)-->  –C(OH)–C(X)–  +  HX        (X2 = halogen; product =
halohydrin)

–C(OH)–C(X)–  --(base)-->  epoxide  +  HX

CH3–CH(OH)–CH2Br  --(OH⁻)-->  methyl-epoxide (propylene oxide)  +  HBr
```

Mechanism: hydroxide removes the proton of the OH group; the resulting alkoxide oxygen attacks the neighbouring carbon from the back side and bromide leaves (an intramolecular SN2), closing the three-membered ring.

Reactions of epoxides: ring opening

Acid-catalysed epoxide opening

Epoxides are cleaved by treatment with acid just as other ethers are, but under milder conditions because of the strong ring strain.

```
epoxide  --(H⁺, Nu–H)-->  H–C(H)(Nu⁺H)–C(H)(H)–OH  --(–H⁺)-->  Nu–CH2–CH2–OH
```

Mechanism:

1. Protonation of the epoxide oxygen by H^+.
2. Nu–H attacks a ring carbon from the back side, opening the ring and giving $Nu^+(H)$–C–C–OH.

3. Loss of H^+ (taken by another Nu–H, which becomes Nu^+H_2) gives Nu–C–C–OH.

Examples:

```
1,2-epoxycyclohexane  --(H3O+)-->  protonated epoxide  --(H2O attacks from
the back)-->  cyclohexane-1,2-diol (trans)  +  H3O+

1,2-epoxycyclohexane  --(HX, anhydrous, ether)-->  2-halocyclohexanol   (OH
on one carbon, X on the adjacent carbon)
```

Regiochemistry

- When both epoxide carbon atoms are either 1° or 2°, the nucleophile attacks mainly the less highly substituted site, hence an **SN2-like** result.
- When one of the epoxide carbon atoms is tertiary, nucleophilic attack occurs primarily at the more highly substituted site, hence an **SN1-like** result.

Examples drawn in the notes (H^+, then CH_3OH, then $-H^+$). The products below follow the rule stated in the notes, because the skeleton drawings are small:

```
1°/2° epoxide (propylene oxide):   CH3–CH(–O–)CH2  --(H+, CH3OH)-->  CH3–
CH(OH)–CH2–OCH3
                                    (attack at the less substituted CH2)

3° epoxide (2,2-dimethyloxirane):  (CH3)2C(–O–)CH2  --(H+, CH3OH)-->  HO–CH2–
C(CH3)2–OCH3
                                    (attack at the tertiary carbon)
```

Base-catalysed epoxide opening

The strain of the epoxide ring causes it to react with hydroxide ion.

```
cyclohexane ring carbon bearing the epoxide (C–O–CH2)  --(OH-)-->  ring
C(O-)–CH2–OH  --(H2O)-->  ring C(OH)–CH2–OH
```

They can react with Grignard reagents as well:

```
R–MgBr  +  ethylene oxide  -->  R–CH2CH2–O- +MgBr  --(H3O+)-->  R–CH2CH2–OH
```

Base-catalysed epoxide opening is typical of an **SN2 reaction**, in which attack of the nucleophile takes place at the less hindered epoxide carbon atom.

```
epoxide on a cyclohexane ring carbon  --(CH3CH2OH / -OCH2CH3)-->  ring C(OH)-CH2-OCH2CH3
```

The ethoxide attacks the CH_2 carbon (the less hindered one), so the OH stays on the ring carbon.

5. Aldehydes and ketones

Preparation, properties and reaction mechanisms of: (1) aldehydes, (2) ketones, (3) α,β-unsaturated carbonyl compounds.

```
    O                  O
    ||                 ||
R-C-H              R-C-R'
aldehyde            ketone
```

Nomenclature of aldehydes

The IUPAC name of an aldehyde is obtained by replacing the final "e" in the name of the corresponding alkane with **"al"**.

Structure	Name
H–CHO	Methanal
CH_3–CHO	Ethanal (acetaldehyde)
$CH_3CH_2CH_2$–CHO	Butanal (butyraldehyde)
CH_3CH_2–CH(Cl)–CHO	2-Chlorobutanal
$(CH_3)_2$CH–CHO	2-Methylpropanal
OHC–$CH_2CH_2CH_2$–CHO	Pentanedial

If the aldehyde is attached to a ring, it is named by adding **carbaldehyde** to the name of the ring:

- Cyclohexane ring bearing –CHO: cyclohexanecarbaldehyde
- Benzene ring bearing –CHO: benzenecarbaldehyde (benzaldehyde)

Nomenclature of ketones

The systematic name of a ketone is obtained by replacing the "e" at the end of the name of the corresponding alkane with **"one"**. The carbon atoms are numbered from the end that gives the carbonyl carbon the smaller number.

Structure	Name
CH_3–CO–CH_3	Propanone
CH_3CH_2–CO–CH_3	Butanone (2-butanone)
Skeleton drawing, 6-carbon chain with a methyl branch	5-Methyl-2-hexanone
Six-membered ring with C=O	Cyclohexanone
CH_3–CO–CH_2–CO–CH_3	2,4-Pentanedione
C_6H_5–CO–CH_3	Methyl phenyl ketone / phenylethanone (acetophenone)

Relative reactivities of carbonyl compounds

The carbonyl group is polar because oxygen is more electronegative than carbon. Thus the carbonyl carbon is electron deficient and carries a partial positive charge (δ+). It is therefore an **electrophile** and reacts with nucleophiles.

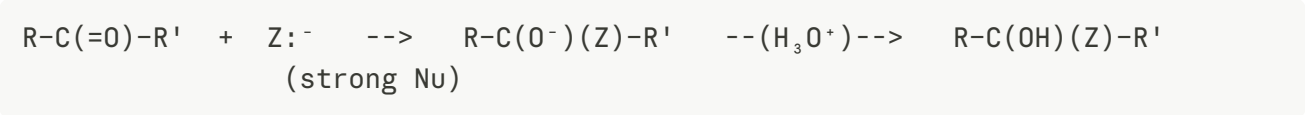
```
   Oδ⁻
   ‖
R–Cδ⁺–R(H)    ←  Nu:⁻ attacks the carbonyl carbon
```

An aldehyde has a greater δ+ on its carbonyl carbon than a ketone, because H is more electron-withdrawing than alkyl.

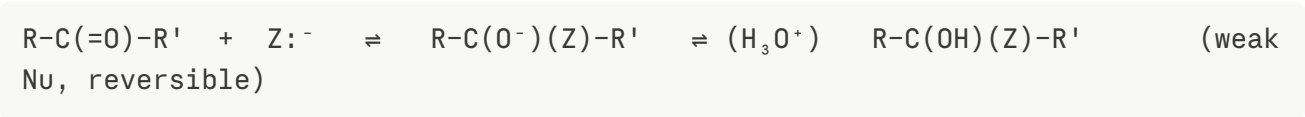
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Most reactive:  H–C(=O)–H   >   R–C(=O)–H   >   R–C(=O)–R'   :least reactive
```

Reactions of aldehydes and ketones

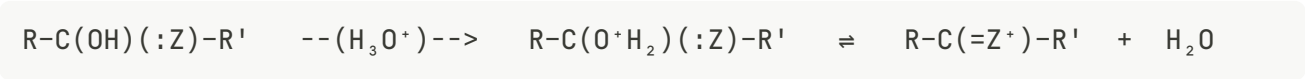
Aldehydes and ketones react with nucleophiles to form addition products.



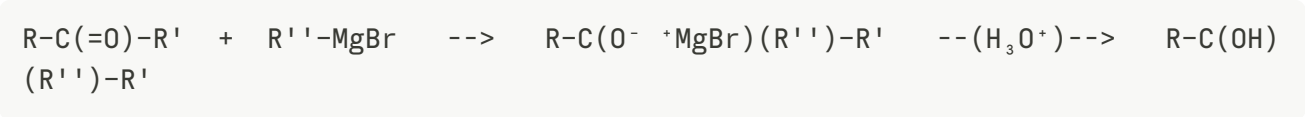
- The reaction above is an example of an **irreversible nucleophilic addition**, because the nucleophile is a strong base and the tetrahedral compound does not have a group that can be eliminated.
- On the other hand, if the nucleophile is a weak base, the reaction will be a **reversible nucleophilic addition**, because the tetrahedral compound can eliminate the weak base and revert to the starting material (Z can be eliminated).



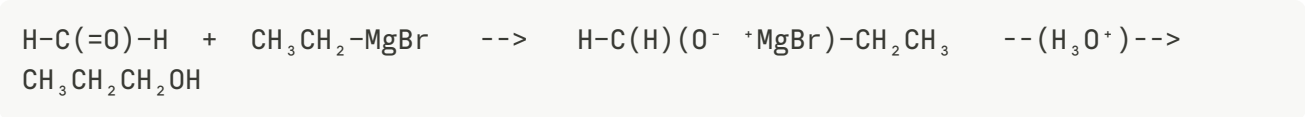
- If the attacking atom of the nucleophile has a lone pair and there is enough acid in the solution to protonate the –OH group, water can be eliminated from the addition product. Thus the fate of the tetrahedral product depends on the identity of Z. This is called a **nucleophilic addition–elimination reaction**.



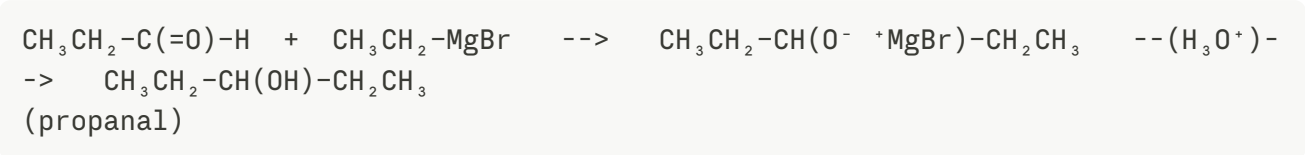
1. Reaction with Grignard reagents



- **Formaldehyde gives a primary alcohol:**



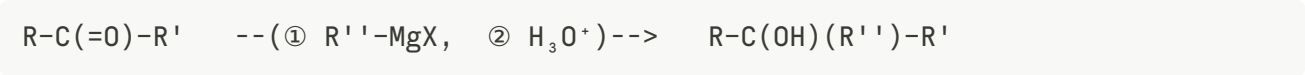
- **An aldehyde other than formaldehyde gives a secondary (2°) alcohol:**



- **A ketone gives a tertiary alcohol:**



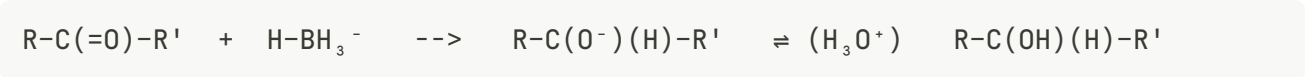
The reaction can generally be written as:



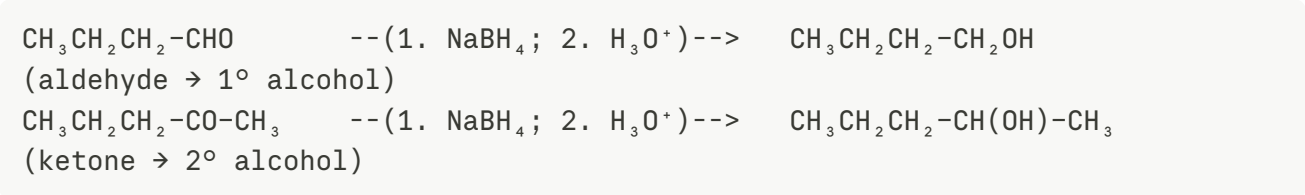
The order of numbering means that the acid is added after the Grignard reagent has reacted.

2. Reaction with hydride ion

Aldehydes and ketones react with a hydride ion to form a nucleophilic addition product.

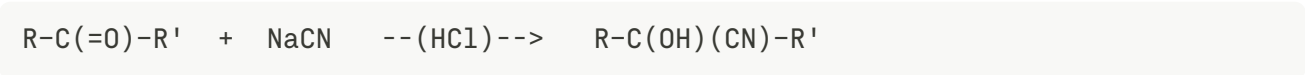


In the presence of a hydride source (such as $NaBH_4$), aldehydes and ketones are generally reduced to primary and secondary alcohols respectively:

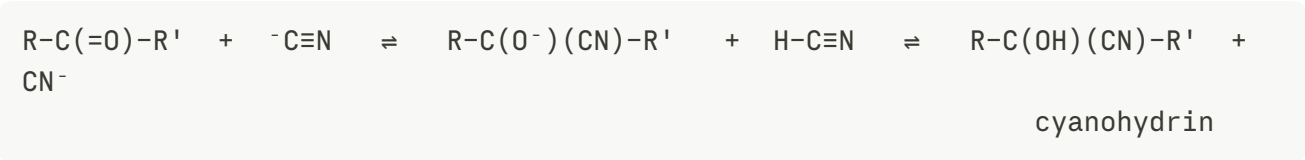


3. Reaction with hydrogen cyanide

An aldehyde or ketone reacts with HCN to form a **cyanohydrin**.



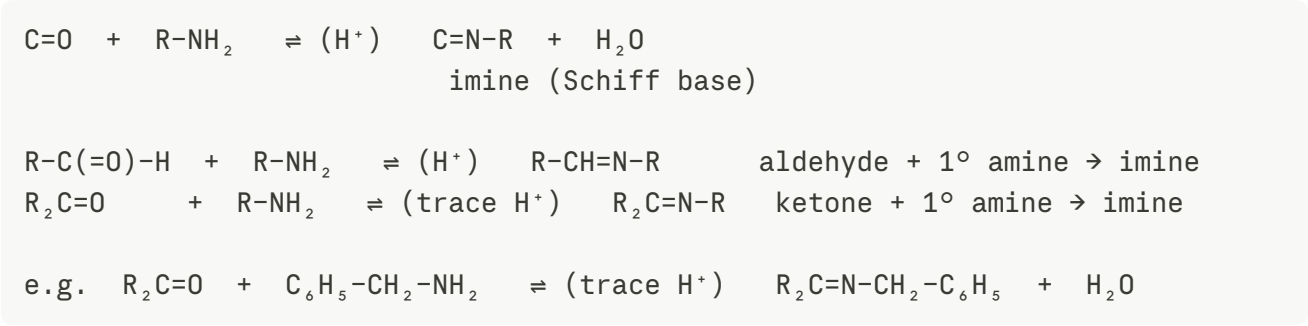
Mechanism:



In basic solution, a cyanohydrin is converted back to the carbonyl compound, for example cyclohexanone cyanohydrin ⇌ (HO^-/H_2O) ⇌ alkoxide ⇌ cyclohexanone + CN^-.

4. Reaction with amines and amine derivatives

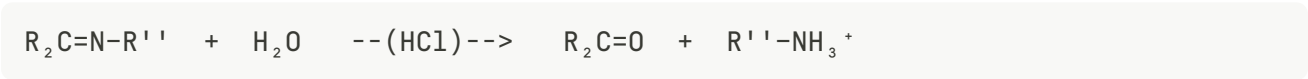
An aldehyde or a ketone reacts with a primary amine to form an **imine (Schiff base)**.



Mechanism (B: = any species that can remove a proton, H–B⁺ = any species that can donate a proton):

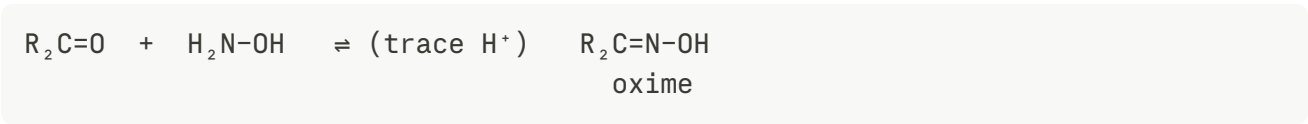
1. The amine nitrogen attacks the carbonyl carbon: R(R′)C(O⁻)–N⁺H₂R″.
2. Proton transfers give the carbinolamine R(R′)C(OH)–NHR″.
3. The –OH is protonated (H–B⁺), giving R(R′)C(O⁺H₂)–NHR″.
4. Loss of water gives the iminium ion R(R′)C=N⁺HR″.
5. Deprotonation by B: gives the imine R(R′)C=N–R″ (and H–B⁺).

In acidic medium, the imine is hydrolysed back to the carbonyl compound:

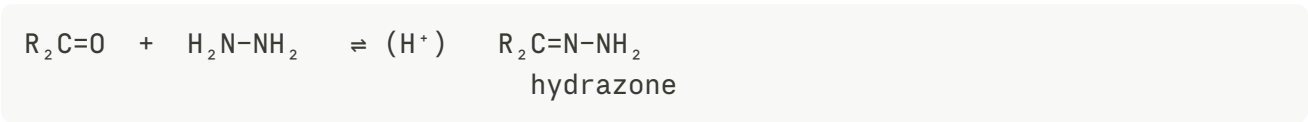


Formation of imine derivatives

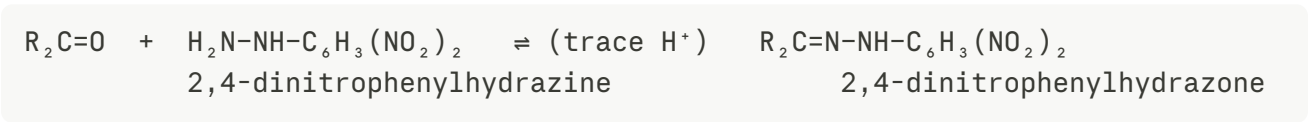
- An aldehyde or a ketone reacts with hydroxylamine to give an **oxime**:



- They also react with hydrazine to give a **hydrazone**:



- With 2,4-dinitrophenylhydrazine:



- *(Boxed and marked "Ignore" in your notes.)* Aldehydes and ketones react with semicarbazide to give a semicarbazone:

